CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review:	**Review–2**	**Date of Review:**	**3 October 2026**
Project Title:	**Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators**		
Cohort:	_____	**Project Guide:**	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

Student Details:

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Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

- **Baseline CNN & Affine INT8 Quantization:** Trained 4-conv baseline CNN on Fashion-MNIST (84.69% exact baseline accuracy). Developed post-training INT8 quantization engine (unsigned 8-bit activations [0, 255], sign-magnitude weights) matching unsigned hardware multipliers.
- **Single-Pass Sensitivity Formulation (S1–S5):** Formulated candidate sensitivity metrics. Extracted first-order loss gradients via TensorFlow GradientTape on a 256-sample calibration batch in 0.2s. Proved conv1 is critical (S=1.000) while conv4 is highly robust (S=0.014).
- **TRUNC=4 Sweep & Noise Floor Discovery:** Executed 12-cell validation sweep across 10,000 test images. Measured accuracy drops fell within ±0.1% to 0.3%, identifying the CNN resilience boundary and revealing an initial statistical noise floor (formal NO-GO at TRUNC=4).
- **TRUNC=6 Scaling & Signal Amplification:** Scaled truncation depth to T=6 (dropping 21 of 64 partial products, amplifying NMED by 6.6x to 0.001234, saving 32.5% area). Proof-of-concept tests demonstrated sensitivity assignment achieved 85.27% accuracy, recovering 81.5% of accuracy lost by uniform approximation.
- **Spearman Correlation & Formal Gate Clearance:** Proposed combined metric S3 achieved Spearman |rho| = 0.80 on TRUNC=6, clearing the pre-defined |rho| >= 0.70 GO gate and validating the core research hypothesis.

Signature of Project Guide: _____ **Date:** _____